

Ezer Ayari

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Professor Dru Macasieb
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Backend Reflection: Database Integration & Server-Side Logic

This week transformed my website from a front-end application into a true full-stack web application. Before this assignment, any information entered into a form would disappear after the page refreshed. With the addition of PostgreSQL, Prisma, and Next.js Server Actions, user data is now permanently stored and can be retrieved whenever needed.

When a user clicks the Submit button on the contact form, the browser sends the form data to a Next.js Server Action running on the server. The Server Action validates the input and uses Prisma, an Object Relational Mapper (ORM), to communicate with the PostgreSQL database. Prisma converts the JavaScript request into SQL queries and inserts the new record into the Message table. The database stores the information permanently. When the user visits the messages page, a Server Component executes a database query through Prisma, retrieves the stored records, and renders the results as HTML before sending the page back to the browser. This process includes the browser, Next.js application, Server Action, Prisma ORM, PostgreSQL database, and Server Component working together.

The modern Node.js and serverless architecture used in this assignment shares many similarities with traditional PHP and ASP.NET applications. Both approaches receive user requests, process server-side logic, communicate with databases, and return results to users. However, modern frameworks such as Next.js integrate client and server functionality within a single project and simplify deployment through platforms such as Vercel. Traditional PHP and ASP.NET applications typically required dedicated web servers and more manual configuration.

One major advantage of having a backend is data persistence. A front-end application alone cannot securely store and manage information across multiple sessions and users. By introducing a database and server-side logic, the application can save user messages, retrieve historical records, and support future features such as user authentication, analytics, and administrative dashboards. This assignment provided a practical understanding of how modern web applications process, store, and display information through a complete end-to-end workflow.

References

Prisma. (2026). Prisma ORM documentation. <https://www.prisma.io/docs>

Vercel. (2026). Environment variables documentation.
<https://vercel.com/docs/environment-variables>

PostgreSQL Global Development Group. (2026). PostgreSQL documentation.
<https://www.postgresql.org/docs/>